Climate.

Climate	Weather
• Sum total of weather conditions and variations over a large area for a long period of time. (For more than 30 yrs).	• State of atmosphere over an area at any point of time. (Day to day phenomena).

- Elements of climate and weather remains same.

• It includes → 1) Temperature.
2) Wind.
3) Precipitation
4) Atmospheric Pressure.
5) Humidity.

- The difference between climate and weather is of time period.

⇒ **Climate of India:**

• The climate of India has a Monsoon type.

- Monsoon → derived from an Arabic word 'Mausim' which means 'Season'.
- In Asia Monsoon type of climate is found in south ~~Asa~~ Asia and South - East Asia.
- It includes Bangladesh, Pakistan, Malysia, Thailand, Vietnam, Singapore, etc.
- In Monsoon type of climate there are regional variations. Temperature and precipitation varies from state to state.

e.g.:- In Summers, Rajasthan → 5°C.
Summer, Kashmir → 20-30°C.
- In Winter, Drass, Ladakh → -45°C.
- In Winter, Thiruvananthanampur → 25°C.

e.g.2) Day and Night temperature:-
Rajasthan → Day → 50°C
Night → 15°C
] → Huge difference

- Coastal areas (Kerala) → hardly any difference in temperatures in day and night.

⇒. Precipitation varies in the form, the amount of rainfall and seasional distribution.

e.g) Upper parts of ~~them~~ Himalayas → Snowfall
Rest of the country → Rainfall.

e.g) Meghalaya → more than 400cm rainfall
Ladakh → less than 10cm.

e.g. Most of the country → June - Sept. (Rainfall)
Some parts of T.N → Oct - Nov
Kashmir → Winter rainfall.

- Due to these regional variations people have different living, culture, clothing, the food they eat and the kind of house they live in.

(i) Houses in Terai, Goa, Mangalore → have sloping roofs → because of heavy rainfall.

. Houses in Rajasthan → Thick walls ~~and flat~~ and flat Roof.

Thick walls ↓ To prevent heat from entering into the house

Flat Roof ↓ To preserve water (also provides cooling effect).

⇒ Factors affecting climate (Climatic control)

① Latitude. ② Altitude. ③ Distance from sea.

④ Ocean currents. ⑤ Relief features.

⑥ Presure ~~of~~ and wind system.

① Latitude :-

- Temperature decreases from equator towards the poles.

- Temperature at equator is more → Tropical or Equitorial type of climate.
- Temperature at poles is less → ~~Temperature~~ Temprate and frigid type of climate.

• India → tropic of cancer divides it almost into 2 equal halves.

① Part laying south of tropic of cancer → Tropical climate.

② Part lying north of tropic of cancer → Sub-tropical

• India has both tropical and sub-tropical climate.

② Altitude :- It is height above the sea level.

• As we move up, gases become less dense and temperature decreases.

- Thats why hilly areas experience milder summer.

e.g. Himalayan region and Iran are present at same latitude but different altitudes, thus experience different climates.

- Himalayas present ~~at~~ in the north prevent cold winds from central Asia to enter into India, thus India experiences milder winters.

③ <u>Distance from the sea:</u> (Continentality)

- Areas present near the sea ~~&~~ → Moderate Climate.
- Areas present far from the sea → Extreme Climate.

→ It occurs because of land breeze and sea breeze e.g. Mumbai, Keralam → Near the sea → Moderate Climate.

Delhi, U.P → far from the sea → Extreme Climate.

④ <u>Ocean Currents.</u>

- It is the continuous movement of ~~water~~ ocean water in a perticular direction. It is influenced due to gravity, wind or rotation of earth.

→ <u>Two Types</u>-

1) Warm Ocean Currents. (Raise the temperature of place)
2) Cold Ocean Currents. (Drop the temperation of place).

→ El-Nino (Warm Ocean Currents)

La-Nina (Cold Ocean Current).

- Ocean currents influence the location of desert. Cold ocean currents don't have moisture and thus bring little to no rainfall resulting in the formation of deserts in western margins of continents.

⑤ <u>Relief</u>:- Physical features of earth.

- Different relief features have different type of climate.

e.g. Mountain → Windward side → Rainfall occurs here.
↳ Leeward side → Remain dry.

- Western Ghat → West side → Windward side → Rainfall
↳ East side → Leeward → ~~Dry~~ Dry side

⑥ <u>Pressure and Wind System</u> :-

- Areas where pressure is high → Air sinks, it becomes warmer and drier → No rainfall. ~~Cli~~

→ Areas where pressure is low → Air rises, cools and brings rainfall → Moist Climate.

⇒ Winds ~~travel~~ travel from high pressure to low pressure.

⇒ Pressure and Wind System (India):-

- 3 wind conditions that influence the climate of India:-

1) Pressure and Surface Wind.

2) Jet streams (Upper air circulation)

3) Western ~~disturbances~~ disturbances.

1) Pressure and surface Winds:-

- - - - →

Winter	Summer
• High pressure is created over north of Himalayas and cold winds blow from this region to low pressure area over Indian ocean to south.	• High pressure develops over Indian ocean and low pressure over north of Himalayas thus winds move from ocean to the mainland (Monsoon).
• They move over land, thus don't carry any rainfall/moisture.	• These winds carry moisture and brings rainfall in the Indian

2) Jet Streams :→

⇒ • High speed winds that blow above the surface of ~~*~~ the earth (10-15 km).

• Speed - 110 km/hr - 184 km/hr.

- These winds influence monsoon winds and western disturbances

3) <u>Western Disturbances:</u>

- They are the winds that originate from ~~Mediterranean~~ Mediterranean sea and move eastwards forwards N and N-W. India.

- Bring rainfall and snowfall in the N- and N-W India.

⇒ <u>Features that influence monsoon Winds:-</u>

1) Different heating and cooling of land and water:

- Land and sea heat and cool at different rates

Land → Low Pressure.
Water → High Pressure.

Monsoon winds travel ~~fam~~ from high pressure to low pressure.

2) Inter tropical Convergence Zone (ITCZ):

- It is a low pressure belt near the equator where N-E winds and S-E winds meet.
- During Summer ITCZ shift towards north from equator to India. This will attract the Moonsoon winds from high pressure created over sea to low pressure over the land.

3) Presence of high pressure over Madagascar:

- There are low sun rays during summer in southern Hemisphere which creates high pressure over Madagascar and helps monsoon winds to reach India.

4) Intensly heated tibetan plateau:

- During summer Tibetan plateau gets heated which forms a low pressure belt over it.
- This attracts monsoon winds and help bring monsoon to India.

⇒ Seasons in India-

- There are 4 seasons in India

1) The cold weather season (Winter)
2) The hot weather season (Summer).
3) The advancing monsoon (Rainy season).
4) The retreating monsoon (Transition season).

1) Cold Weather Season:

- It starts from mid november to february.
- December and january are the coldest months
- of north India.
- Temperature decreases from South to North.

e.g In Chennai temperature is 24-25°C while as in Northern plains temperature is 10-15°C.

- Frost is common and slops of himalaya experience snowfall.
- N-E winds blow during winters, dry winds that blow on land and don't bring any moisture, thus no rainfall.
- These winds bring some amount of rainfall as

they move over sea (Bay of bengal) and bring rainfall in some parts of Tamil Nadu.

- During Winter - Western cyclonic disturbances also occur as they bring rainfall and snowfall in North and N-W India.

• Total amount of rainfall during winter is locally c/a Mahavat.

- It is important for growth of rabi crops.

- In peninsular India, → no well defined Winter season because the temp. remains moderate because of the distance from the sea.

2) <u>The Hot Weather Season</u> (Summer)

• Between March - May.

- Sun shifts northwards, temp. in north hemisphere increases and pressure falls.

→ March → 38°C, April - 42°C, May → 45°C.

→ by the end of May, low pressure belt is created over the north India

→ During Summer 'Loo' is blown in the northern India.

Strong, gusty, dry, hot winds and blow day → They are fatal.

→ The season is also characterised by thunderstorm, torrential rains (heavy, rainfall for small period of time), followed by hailstorm.

→ In West Bengal, storms are c/a Kal Baisaki.

→ Towards the end of summer, monsoon showers occur (kerala and karnataka).

- They help in ~~so~~ riping of mangoes - thats why c/a 'mango showers'.

3) <u>The Advancing Monsoon</u>- (Rainy Season)

★- Starts in early June.

- During early June low pressure is created over the Northern hemisphere. This attracts the winds from the southern hemisphere. First the S-E winds travell and once they cross equator, they blow in the S-W direction.
- They are called as ~~S-E~~ S-W monsoon Winds.
- As they blow over the oceans they carry abondand moisture to the Indian subcontinent

and brings rainfall to whole area.

→ These winds are strong and blow at an average velocity of 30 km/hr.

→ These winds cover the whole india in about a month and brings change in the entire weather system of India.

→ As the monsoon starts the windward side of western ghat receives very heavy rainfall. (More than 250cm).

- Deccan ~~Plate~~ Pleatue and parts of M.P. also receives some ~~parts~~ amount of rainfall even though they ~~d~~ lie in rain shadow area.

→ Maximum rainfall is recieved in the N-E par of the country. (Through bay of Bengal branch).

→ Mawsynram (Meghalaya):- Present on the Khasi hills → Receives the maximum rainfall in the world. (Wettest place).

→ Rainfall decreases from E-W Rajasthan and Gujrat recieves scanity rainfall.

→ One of the important phenomena of monsoon is the ~~bra~~ breaks in the rainfall. It has wet and dry spells. It is because of the monsoon trough. When this trough moves over plains, that part recieves the good rainfall. If is moves closer to himalayas, there are dry spells in the plains and rainfall over the foothills of himalayas.

→ Monsoon brings heavy rainfall causing devastating flood and damage to life and property. Because of its incertainity, farming schedule is also disturbed.

4) <u>Retreating Monsoon</u>: Also called as Post-Monsoon Season.

☆→ Starts in October.

- During October November, movement of the sun occurs towards Southern-hemisphere. Low presure belt over Indian sub-continent weakens.
- Gradually, it is replaced by high pressure belt.

- It weakens S-W monsoon winds, start gradually withdrawing from october.

- **Climate:** Dry climate, clear skies, high day temperature, humidity is also high → Oppresive weather → Which is called as 'October heat'.

- Nights are cool and pleasant.

- In the second half of the October → Temperature drops in North-India.

- In early Nov. → low pressure shifts over Bay of Bengal which creates tropical cyclones. They cause destruction in eastern coast of India, also thickly populated areas of Godavari, krishna and kaveri → are also struck with cyclones causing damage to life and property.

→ These cyclones also struck, Odisha, W.B and Bangladesh and also brings rainfall in Coromandal coast.

⇒ **Distribution of Rainfall:-**

- Annual rainfall in India due to monsoon varies from year to year.

→ Variability in rainfall can be seen from state to state:-

1) In Western coast and in North eastern states → more than 400 cm rainfall occurs → Recieves high rainfall. → Can also lead to flood like condition

2) Gujrat, Rajasthan → Recieves scanty rainfall.
↳ Drought like condition occur.

3) Leaward side of W. Ghat, interior of Deccan plateau, Ladakh → Recieves low precipitation.

4) Punjab, Haryana → Less than 60cm rainfall.

5) Rest of the country receives moderate rainfall.

6) Snowfall is restricted to Himalayan regions.

- Thus, the distribution of rainfall is Uneven in the India.